

Classification of Elements and Periodicity

1. State Modern Periodic Law.

Ans: The modern periodic law was proposed by Henry Moseley. Statement:
The physical and chemical properties of elements are periodic functions of their atomic numbers.

2. State Mendeleev's Periodic Law with merits and demerits.

Ans: The physical and chemical properties of elements are periodic functions of their atomic masses.

Merits:

- a. Systematic Arrangement
- b. Prediction of new elements
- c. Correction of atomic masses

Demerits:

- a. Position of hydrogen uncertain
- b. Isotopes not explained
- c. Anomalous pairs present

3. How are the anomalies of Mendeleev's Periodic table removed in modern periodic table?

Ans: Removal in Modern Periodic Table

- Elements arranged by atomic number instead of atomic mass.
- Isotopes placed in same position.
- Anomalous pairs corrected.
- f-block elements placed separately.

Atomic Radii

Atomic radius is the distance from the center of the nucleus to the outermost electron shell of an atom.

In Group: Going down a group, the atomic radius of an atom increases. On going down each element new shell is added on the outermost layer which generally increases the atomic radius.

In Period: Going left to right, there is no shell added to the atom each and every atom has equal shell but the number of electrons keep on increasing that increases the nuclear charge (force of attraction) within atom that's why the atomic size generally decreases.

Ionization Potential (Ionization Energy)

Ionization Potential is the minimum amount of energy required to remove the most loosely bound electron from an isolated gaseous atom.

Types: a. First Ionization Energy: Energy required to remove first electron.
b. Second Ionization Energy: Energy required to remove second electron.

Factors affecting IP:

1. Atomic Size (inversely proportional to IP)
Larger size = Lower IP
Small size = Higher IP
2. Stable configuration
Half-filled and full-filled are more stable and have high IP.
3. Shielding Effect
blocking of nuclear attraction by inner shell electrons

In Group (Top to Bottom): IP decreases as Atomic size increases.

In Period (Left to Right): IP increases

Exceptions: N and O Mg and Al

Electron Affinity

Electron Affinity (EA) is the amount of energy released when an isolated gaseous atom gains an electron

Factors affecting EA:

1. Atomic Size (inversely proportional to EA)
Larger size = Lower EA
Small size = Higher EA
2. Stable configuration
Half-filled and full-filled are more stable
3. Shielding Effect
blocking of nuclear attraction by inner shell electrons

In Group (Top to Bottom): EA decreases as Atomic size increases.

In Period (Left to Right): EA increases

Note: Noble gas and Group 2 element has 0 EA (Almost 0).

Exceptions: F and Cl

Electronegativity

Electronegativity is the ability of an atom in a molecule to attract the shared pair of electrons toward itself.

In Group (Top to Bottom): EN decreases

In Period (Left to Right): EN increases

Priority: $F > O > N$, $Cl > Br > I > C$, $S > P$, H

Q. Ar comes before K in the periodic table, but Ar has a larger relative atomic mass than K. Explain.

Ans: In the modern periodic table, elements are arranged according to increasing atomic number, not relative atomic mass, this happens because atomic mass depends on the isotopes of the element, argon has heavier isotopes in higher proportion which increase its relative atomic mass.

Q. Is a Potassium atom larger, smaller or same size as potassium ion?

Ans: A Potassium atom is larger than a Potassium ion (K^+) because When potassium forms K^+ , it loses one electron. Losing an electron removes the outermost shell and increases the effective attraction between the nucleus and remaining electrons, so the ionic radius decreases.

Note: $K^+ < K \text{ atom} < K^-$ (Almost for all the atoms)

Q. Which of the following would have larger size and why? F or F^- .

Ans: Fluoride ion (F^-) has a larger size than Fluorine (F atom) because F^- has one extra electron, which increases electron–electron repulsion, so its size increases.

Q. Predict which of the following pairs has higher electron affinity and why? O and F.

Ans: Fluorine has higher electron affinity than Oxygen because F is smaller than O, so In F the added electron is closer to the nucleus, so the nucleus attracts it more strongly $\rightarrow$ more energy released $\rightarrow$ higher EA. (small size=high E.A)

Q. Arrange the elements Li, Na, and K in the increasing order of first ionization energy.

Ans: The first ionization energy is the energy required to remove the outermost electron from a neutral atom. All the elements Li, Na and K belongs to Group 1 and IP decreases going down a group due to increase in atomic size. So, $K < Na < Li$ is the order.

Q. Which one N or O has higher electron affinity and why?

Ans: Oxygen has higher electron affinity than Nitrogen because the configuration of N is $1s^2 2s^2 2p^3$ which is half filled i.e. stable but O has $2p^4$ not half filled so Adding one electron completes a half-filled configuration $\rightarrow$ more stable $\rightarrow$ more energy released $\rightarrow$ higher EA. (Better to make configuration fig as well for p orbital)

Q. Which one N or O has higher Ionization Energy and why?

Ans: Nitrogen has higher Ionization energy than Oxygen because the configuration of N is $1s^2 2s^2 2p^3$ which is half filled i.e. stable $\rightarrow$ harder to remove an electron $\rightarrow$ higher IE. But oxygen has $2p^4$ one extra electron in 2p orbital $\rightarrow$ slightly less stable $\rightarrow$ easier to remove an electron $\rightarrow$ lower IE.

(Same for Mg and Al)

Q. Element 'X' lies in Group A and element 'Y' also lies in same group. Explain Why element 'X' shows similar properties with another element 'Y'?

Ans: Element X shows similar properties to element Y in the modern periodic table because both elements possess the same number of valence electrons.

Q. Explain how electronegativity difference between elements helps to predict the nature of bond between two elements.

Ans: Electronegativity difference between elements helps to predict the nature of bond between two elements. In general, large difference in electronegativity results in ionic bonds, while smaller differences result in covalent bond.